Introduction to Random Signal Representations

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Key Concepts

1) Uncertainty present in signals is modeled using tools of probability and statistics.

2) The probability density function completely characterizes a random signal or a scalar random variable.

3) Example probability density functions:

    a) Uniform implies any value within a specified range is equally likely.

b) Gaussian is the familiar bell-shaped probability density function. Described completely by its mean and standard deviation.

c) Many other densities, such as chi-squared, Poisson, F, etc

4) Partial characterization of random variables in terms of mean and variance is often used.

a) The true probability density function may be unknown and impossible to determine.

b) The complete information in the true probability density function may not be necessary for some signal processing purposes.

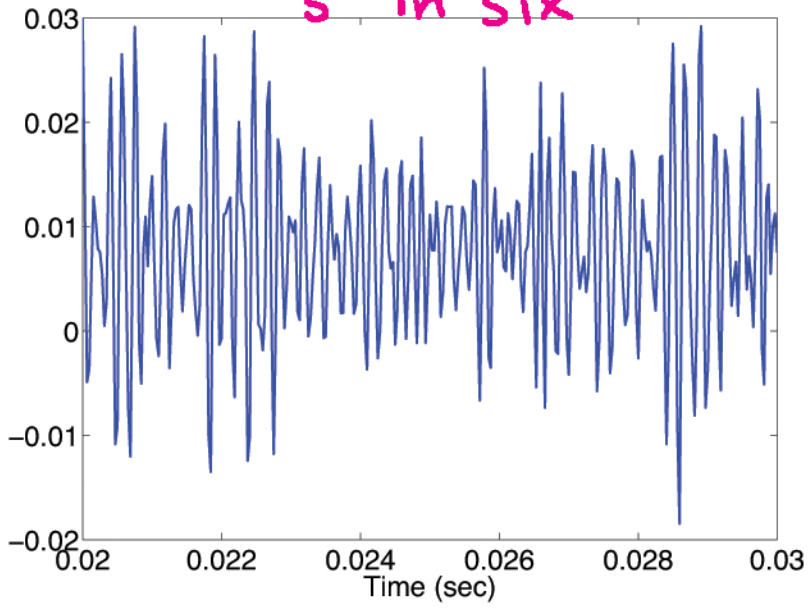
5) Mean is the average value of the random variable over many (an infinite number) of experiments.

6) Variance is the average squared distance of the random variable from the mean over many (an infinite number) of experiments.

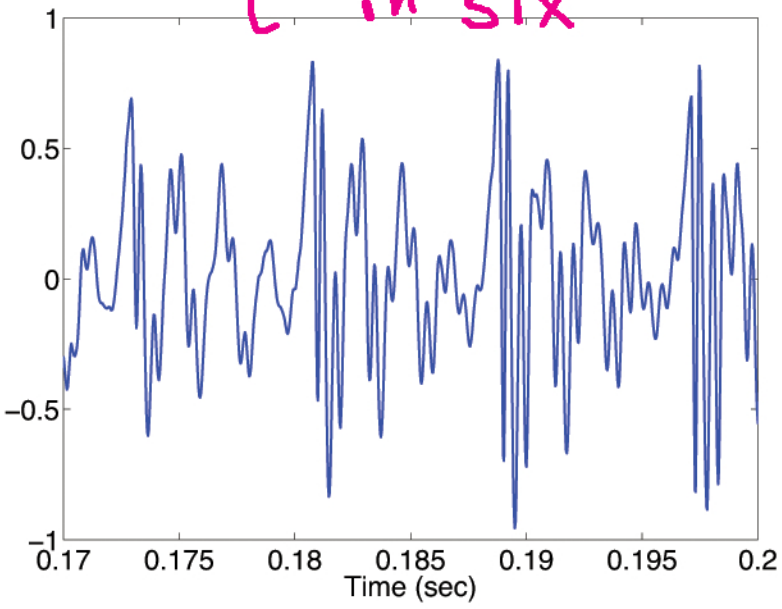
Video Screenshots

Real signals often appear to have random fluctuations

"s" in "six"



"i" in "six"



Use tools of probability and statistics to model uncertainty in a principled manner



probability density function characterizes the possibilities when a waveform is chosen

Start with scalars -

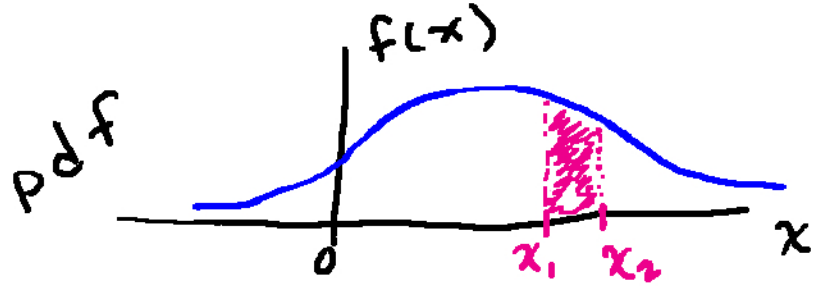
random variable x: symbol representing an unknown outcome

Example:

- binary value $x=1$ or $x=0$

  $\text{Prob}\{x=0\} = 1/2$, $\text{Prob}\{x=1\} = 1/2$ "equally likely"

- continuous value $-\infty < x < \infty$

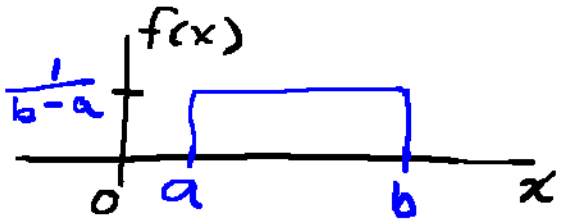


$$\text{Prob}\{x_1 < x < x_2\} = \int_{x_1}^{x_2} f(x)\,dx$$

Common probability density functions

1) Uniform

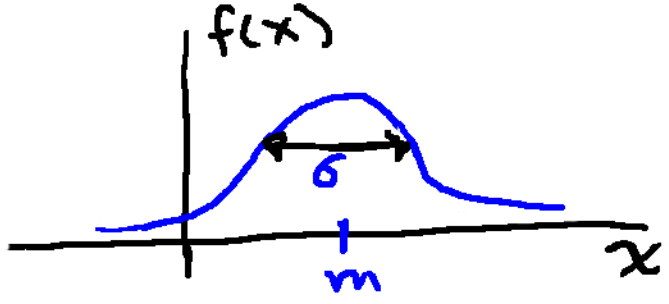
$$f(x) = \begin{cases} \frac{1}{b-a}, & a<x<b \\ 0 & \text{otherwise} \end{cases}$$



equally likely any where in $[a, b]$

$x[n] = A\cos(\omega_0 n + \phi)$ $\quad \phi \sim [0, 2\pi]$

2) Gaussian



$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x-m)^2\right\}$$

m: center of concentration (mean)

σ: tells how concentrated

3) Chi-squared, Poisson, F, etc

Partial Characterization -

1) pdf may be unknown
2) May not need level of detail in pdf

Summary Statistics

1) Mean - on average, what value is observed?
2) Variance - on average, what is the squared distance from the mean?

Expectation: result of "averaging" over an infinite number of experiments $E\{\cdot\}$

1) mean - $m = E\{x\}$
2) variance - $\sigma^2 = E\{(x-m)^2\}$

Formally: $E\{g(x)\} = \int_{-\infty}^{\infty} g(x) f(x) dx$ ($f(x)$: pdf)

So $m = E\{x\} = \int_{-\infty}^{\infty} x f(x) dx$

$$\hat{m} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad \hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N} (x_i - m)^2$$

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